

RailFlow — Redesigning the IRCTC Booking Experience

UX/UI Case Study

Project Overview

Project Name - RailFlow

Project Type - IRCTC Mobile App Redesign

Platform - iOS Mobile Application

Duration - 2–3 Weeks

Role - UX/UI Designer

Tools Used

- [Figma](#)
- [Sharpen.design](#)
- [Iconify](#)
- [Google Fonts \(Inter\)](#)

Introduction

Train travel is one of the most widely used transportation systems in India, and millions of users rely on:

IRCTC Rail Connect

for booking railway tickets.

However, the current booking experience often feels:

- outdated
- visually cluttered
- difficult to navigate
- overwhelming for users

To improve this experience, I redesigned the IRCTC mobile booking flow into a modern and user-friendly product called:

RailFlow.

The goal was to create a cleaner, faster, and more intuitive train booking experience while maintaining the core functionality users already rely on.

Problem Statement

The existing IRCTC booking experience contains several UX challenges:

- information overload
- inconsistent visual hierarchy
- outdated interface patterns
- complicated seat selection
- overwhelming train results
- lengthy booking flow
- limited modern mobile interactions

Users often struggle during:

- train discovery
- seat selection

- passenger information entry
- payment checkout

This redesign focused on simplifying the complete booking experience.

Goal

The main objective of RailFlow was to:

- redesign the IRCTC train booking experience
- modernize the visual interface
- improve usability
- simplify navigation
- reduce cognitive overload
- create a premium mobile-first experience

Research & Inspiration

Before starting the redesign, I explored several modern mobility and booking experiences to understand:

- layout systems
- booking interactions
- mobile usability
- clean information hierarchy

Inspiration Sources

- Uber
- MakeMyTrip
- Airline booking apps
- Modern fintech applications

I also used:

Sharpen.design

to generate UX/UI prompts and explore real-world redesign ideas related to transportation and travel booking systems.

Sharpen.design helped in:

- identifying redesign opportunities
- exploring travel-focused UI concepts
- refining layout ideas
- generating creative booking experiences
- improving overall design direction

This research phase helped shape the final visual identity and UX flow of RailFlow.

Target Users

Primary Users

- students
- office commuters
- tourists
- daily train travelers
- family travelers

Age Group

18–45 years

User Needs

- faster booking

- easy train comparison
- better seat visualization
- secure payments
- mobile-friendly interactions

Design Process

The project followed a structured UX process:

1. Problem Analysis
2. Research & Inspiration
3. User Flow Planning
4. Wireframing
5. UI Exploration
6. Design System Creation
7. High-Fidelity Screen Design
8. Interactive Prototyping

User Flow

The redesigned booking journey includes:

- 1 Search Trains
- 2 Explore Train Results
- 3 Select Seats
- 4 Add Passenger Details
- 5 Make Payment
- 6 Receive Booking Confirmation

The flow was intentionally simplified to reduce friction and improve clarity.

Design System

Primary Color **#2D4BFF**

Used for:

- buttons
- selected states
- active chips
- payment highlights
- navigation states

Typography

Font Family - Inter

Chosen because of:

- readability
- clean modern appearance
- mobile optimization
- strong hierarchy support

Final Screens

Home Screen

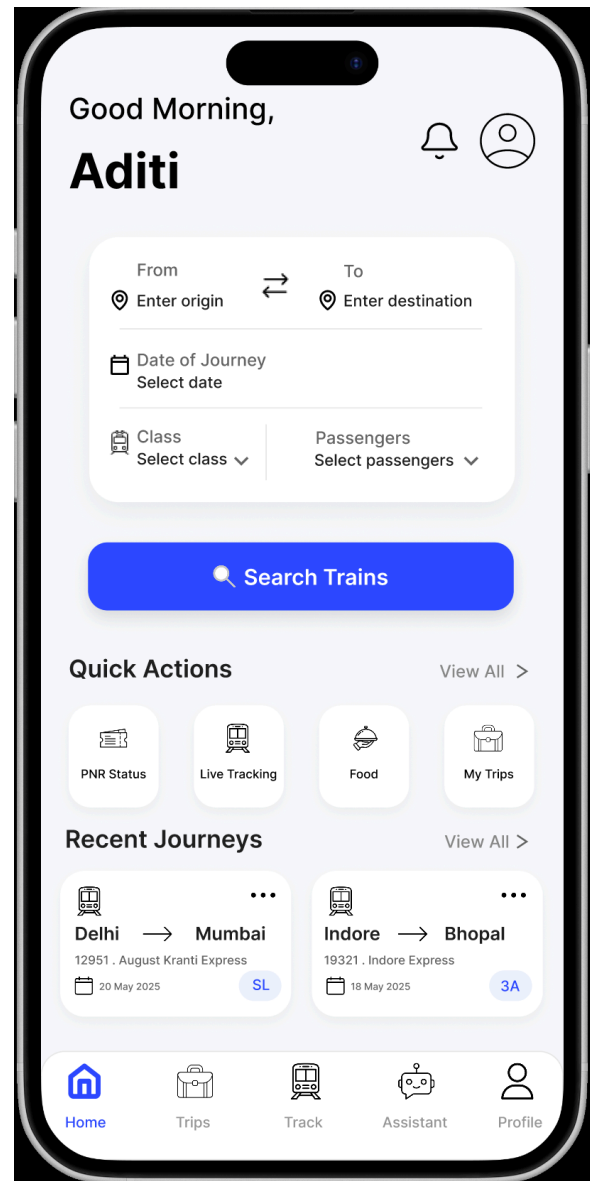
Features

- train booking form
- quick actions

- recent journeys
- bottom navigation

Objective

Allow users to quickly begin the booking process.



2 Search Results Screen

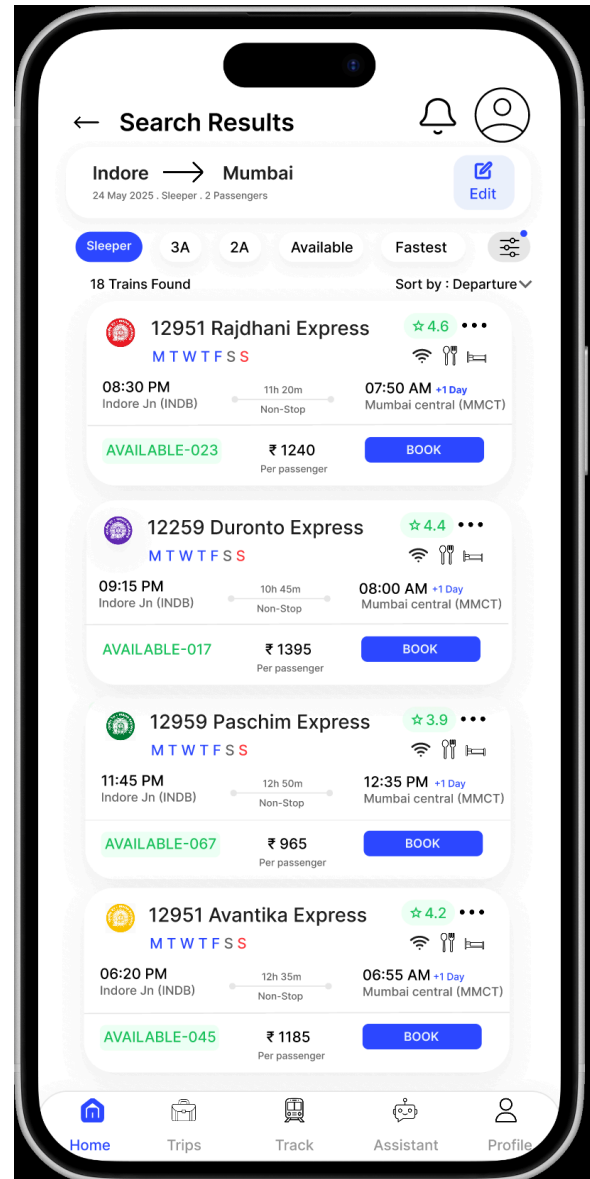
Features

- train filters
- availability indicators
- train comparison cards

- pricing system
- booking CTA

Objective

Simplify train discovery and comparison.



3 Seat Selection Screen

Features

- realistic sleeper coach layout
- berth visualization
- seat legend

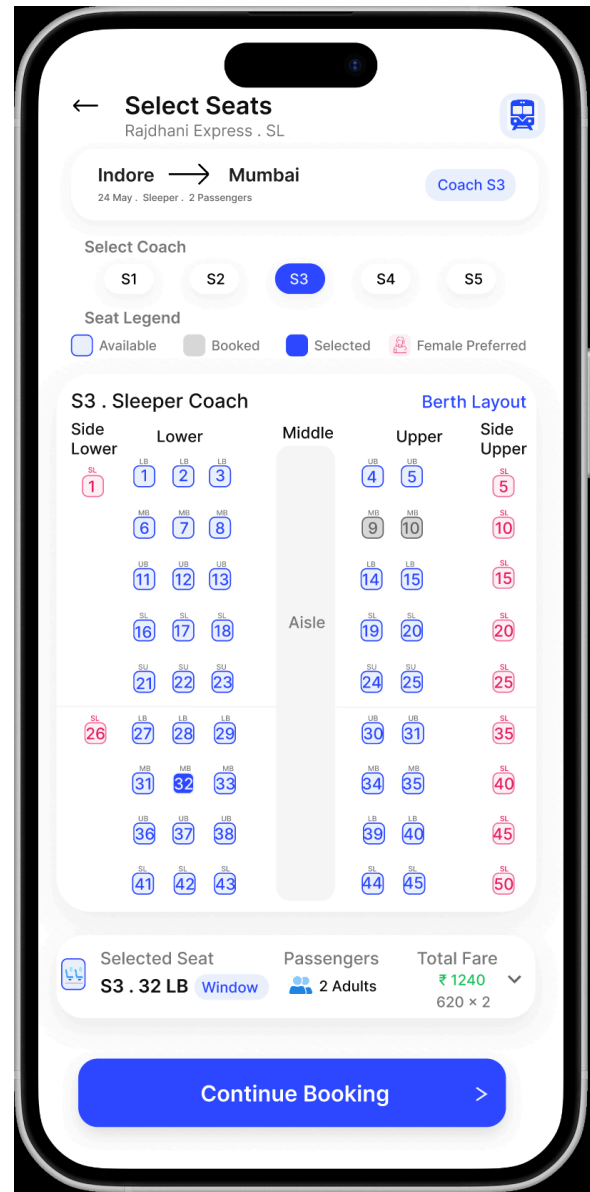
- selected seat summary

Objective

Improve clarity during seat selection.

Highlight

This became one of the strongest screens because of the realistic and interactive berth layout system.



4 Passenger Details Screen

Features

- passenger form fields
- berth preference

- contact information
- fare summary

Objective

Create a clean and easy passenger entry experience.

The mockup shows a mobile app interface for entering passenger details. At the top, there's a title 'Passengers Details' with a back arrow and a train icon. Below it, the train details are 'Rajdhani Express . S3 . 32 LB'. The route is 'Indore → Mumbai' with a date '24 May 2025' and 'Sleeper . 2 Passengers'. A blue button 'S3 . 32 LB' is on the right. The main section is 'Passenger 1' with fields for 'Full Name', 'Age', 'Gender', 'Berth Preference', 'Mobile Number', and 'Email Address (Optional)'. Each field has a placeholder text and an icon. Below the form is a blue button '+ Add Passenger'. At the bottom, there's a fare summary table and a large blue button 'Proceed to Payment' with a right arrow.

Passenger 1	
Full Name	Enter full name
Age	Enter age
Gender	Select gender
Berth Preference	Select preference
Mobile Number	Enter mobile number
Email Address (Optional)	Enter email address

Base Fare	₹ 1100
Taxes	₹ 80
Convenience Fee	₹ 60
Total Amount	₹ 1240

5 Payment Screen

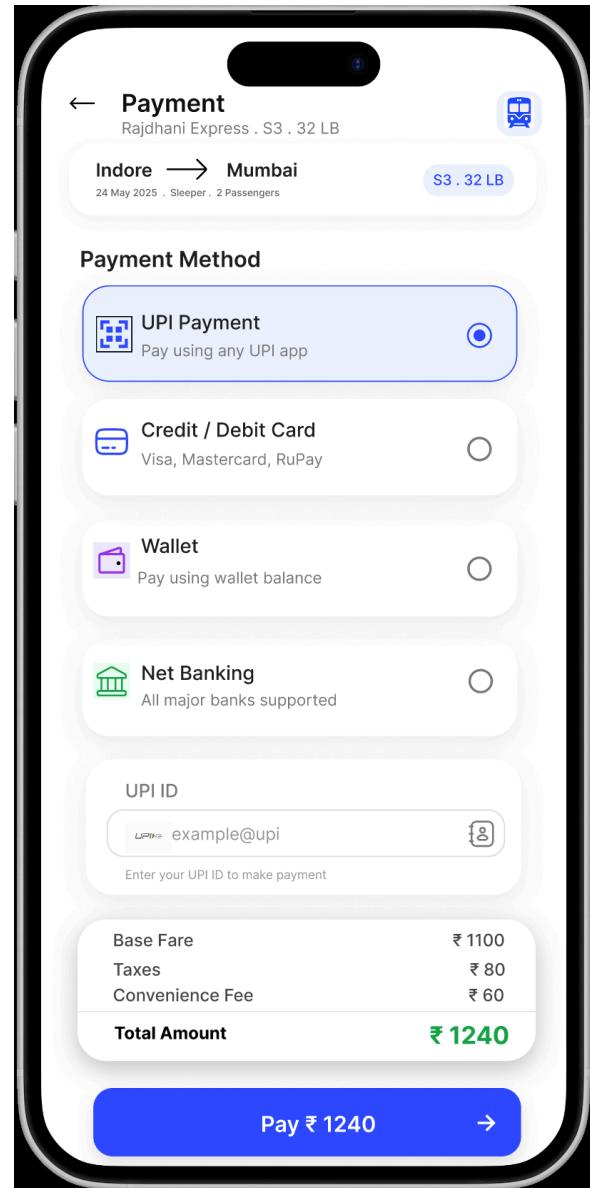
Features

- multiple payment methods
- UPI integration
- fare breakdown

- secure checkout UI

Objective

Deliver a fast and trustworthy payment experience.



6 Booking Success Screen

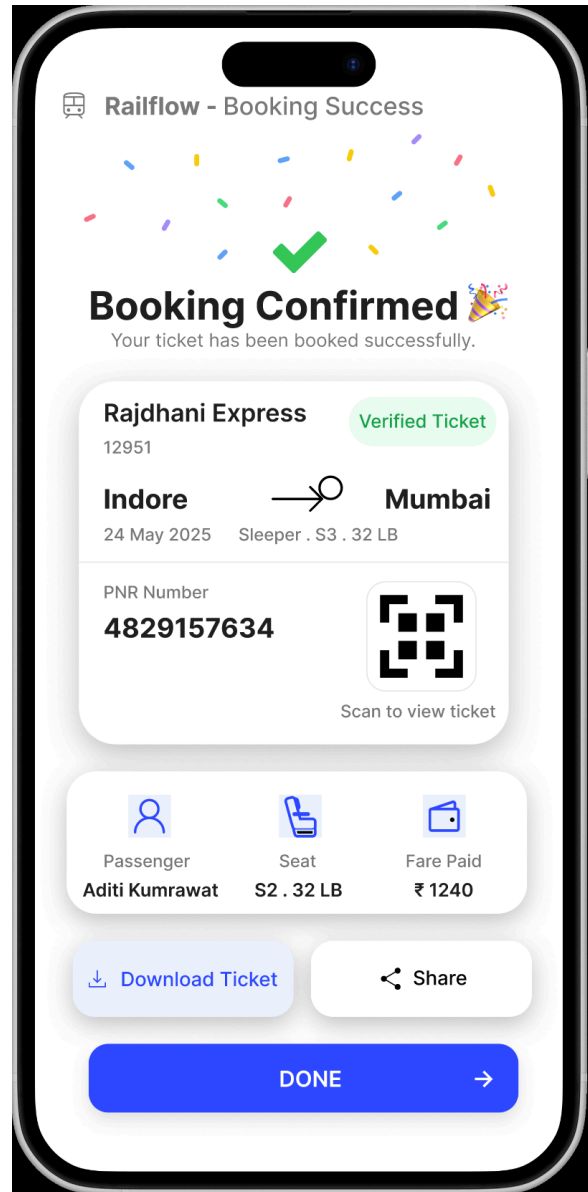
Features

- booking confirmation state
- digital ticket card

- QR verification
- download & share options

Objective

Create a satisfying end to the booking journey.



✨ UI Style

The interface follows:

- minimal UI principles
- soft shadows
- rounded corners
- spacious layouts

- premium fintech-inspired aesthetics

The design language was inspired by:

- modern iOS apps
- Uber
- Apple Wallet
- fintech interfaces

Prototype

An interactive prototype was created in:

Figma Prototype

link : <https://www.figma.com/proto/DStGKvnxvu2Jd4Jtmld21D/Food-Design?node-id=148-3&p=f&t=2u9WjGaFtRNnpOjN-1&scaling=scale-down&content-scaling=fixed&page-id=148%3A2&starting-point-node-id=148%3A3>

using:

- Smart Animate
- screen transitions
- clickable navigation
- interactive buttons

Key Improvements Over IRCTC

Compared to the existing IRCTC experience, RailFlow improves:

- ✓ readability
- ✓ train comparison clarity
- ✓ seat visualization
- ✓ navigation flow
- ✓ visual consistency

- ✓ payment accessibility
- ✓ mobile-first interactions



Challenges Faced

1. Information Density

Balancing large amounts of railway information without cluttering the interface.

2. Seat Layout System

Designing a realistic berth layout while maintaining visual cleanliness.

3. Consistency

Maintaining consistent spacing, shadows, typography, and reusable components across all screens.



Key Learnings

Through this redesign project, I learned:

- UX problem solving
 - design consistency
 - scalable design systems
 - mobile-first interaction design
 - information hierarchy
 - realistic product flow creation
 - interactive prototyping in Figma
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Final Outcome

RailFlow successfully transformed the traditional IRCTC booking experience into a:

- cleaner
- faster
- modern
- user-friendly mobile experience

The final project includes:

- ✓ 6 High-Fidelity Screens
 - ✓ Complete Booking Flow
 - ✓ Interactive Prototype
 - ✓ Consistent Design System
 - ✓ Modern Mobile UI Experience
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Future Improvements

Possible future additions:

- dark mode
 - AI train recommendations
 - live train tracking
 - voice-assisted booking
 - smart delay predictions
 - waitlist prediction system
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Conclusion

RailFlow was designed to reimagine the IRCTC booking experience with a stronger focus on:

- usability
- simplicity
- modern interaction design

- visual clarity

This project demonstrates:

- end-to-end product thinking
- UX strategy
- UI consistency
- realistic mobile workflows
- interactive design systems

The redesign helped strengthen my understanding of:

- user-centered design
- mobile UX
- interaction design
- design systems
- real-world product thinking.